

Cyclistic Bike-Share Data Analysis

English Portfolio Report - Google Data Analytics Capstone
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Executive Summary

This English report summarizes an end-to-end Cyclistic bike-share analysis based on the original Indonesian project report. The analysis compares how casual riders and annual members use bikes differently, then translates the findings into marketing recommendations for annual membership conversion.

The dataset covers one full year of historical trip data from July 2024 to June 2025. The analysis was validated across Excel / Power Query, SQL, Python, and RStudio.

Metric	Value
Total trips after cleaning	5,590,841
Casual rider trips	2,049,922
Annual member trips	3,540,919
Avg. ride length - casual	19.96 minutes
Avg. ride length - member	11.83 minutes
Overall avg. ride length	14.81 minutes

Business Questions

1. How do annual members and casual riders use Cyclistic bikes differently?
2. Why would casual riders buy an annual Cyclistic membership?
3. How can Cyclistic use digital media to influence casual riders to become annual members?

Methodology

The workflow followed Ask, Prepare, Process, Transform, Analyze, Share, and Act. Cleaning included duplicate removal, blank row removal, datetime conversion, ride length calculation, and filtering abnormal rides above 1,440 minutes / 24 hours. Derived fields included hour, day of week, month, ride_length, and category_ride_length.

Key Findings

- Members take more trips overall and show stronger weekday routine behavior.
- Members are more active during commuting windows around 07:00-09:00 and 16:00-18:00.
- Casual riders are more weekend- and season-oriented, especially from May to September.
- Casual riders have longer average ride durations than members on every day of the week.
- Casual riders are more represented in long-duration trips above 30 minutes, creating a conversion opportunity.

Recommendations

1. Launch weekend and summer conversion campaigns during peak casual rider activity.
2. Use time-and-duration-based targeting after long rides, such as: Riding more than 20 minutes? Save more with membership.
3. Use behavior-based digital marketing around leisure areas, parks, tourist spots, and weekend routes.
4. Add cost-saving comparisons to show when membership becomes financially more attractive than repeated single rides.

Limitations and Future Improvements

The dataset does not include unique user identifiers, fare data, demographic data, or weather data. Future work should add rider-level identifiers, payment data, station-level analysis, weather enrichment, predictive conversion modeling, and an interactive Power BI dashboard.

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